Gleyic Lithosols

Gleyic Lithosols are very shallow, poorly drained soils developed over hard rock or compact stony parent material, characterized by gleyic features mottling, greyish-blue colors, and iron concretions caused by prolonged water saturation and reduction.

The prefix *"Gleyic"* denotes the presence of periodic or permanent waterlogging within the shallow soil layer. These soils typically occur in **depressions, lower slopes, or footslope zones** where drainage is restricted by the impermeable rock or compact sublayers.

Because of their shallow depth and poor drainage, they have limited agricultural potential and are often used for grazing or natural vegetation.

Drainage:

- **Wet season:** Poorly drained; waterlogging is common due to slow infiltration and the presence of an impermeable rock or compacted horizon.
- **Dry season:** The upper few centimeters may dry and crack slightly but the sub-surface remains moist or saturated for long periods.
- **Landscape:** Found in low-lying areas, valley bottoms, or footslopes where water accumulates; sometimes on shallow depressions in rocky terrain.

Depth:

- Very shallow, typically **less than 25–40 cm** of fine earth before reaching rock or a cemented horizon.
- **Rooting depth** is extremely restricted; roots are shallow and often affected by anaerobic conditions.

Workability:

- **When wet:** Sticky and plastic, easily compacted; tillage very difficult or impossible due to waterlogging.
- **When dry:** Surface may harden and crack; shallow depth limits effective cultivation.
- Generally unsuitable for mechanized farming; hand tools may be used for minimal surface operations.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (5–15 cmol(+)/kg), depending on parent material and organic content.
- **Base saturation:** Variable, often low to moderate (<50%), influenced by drainage and leaching.
- **Organic matter:** May accumulate near the surface due to slow decomposition under anaerobic conditions (1.5–3%).
- **Nutrients:** Nitrogen and phosphorus are typically low; iron and manganese may be excessive and toxic to sensitive crops during waterlogging.

pH:

- Slightly acidic to neutral (5.5–7.0); acidity increases with prolonged water stagnation and leaching.

Management:

- Improve drainage through shallow ditches or raised beds to reduce waterlogging.
- Use cover crops and grasses tolerant to periodic flooding (rice, Napier grass, vetiver).
- Incorporate organic matter to enhance aeration and nutrient availability.
- Avoid compaction or cultivation during wet conditions.
- Where drainage cannot be improved, the land is best suited for pasture, wetland forestry, or as wildlife and conservation zones.